Algorithm

IstDefinition:

Sequence of steps that can be taken to solve a problem

2nd Definition:

The step by step series of activities performed in a sequence to solve a problem

Better Definition:

A precise sequence of a limited number of unambiguous, executable steps that terminates in the form of a solution

Algorithm

- Algorithm is a step-by-step procedure, which defines a set of instructions to be executed in a certain order to gel the desired output. Algorithms are generally created independent of underlying languages, i.e. an algorithm can be implemented in more than one programming language.

- From the data structure point of view, following are some important categories of algorithms —
- **Search - Algorithm tu search an item in a data structure.**
- **Sort - Algorithm to sort items in a certain order.**
- **Insert - Algorithm to insert item in a data structure.**
- **Update — Algorithm to update an existing item in a data structure.**
- **Delete — Algorithm to delete an existing item from a data structure.**

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Characteristics of an Algorithm

- An algorithm should have the following characteristics -
- **Unambiguous - Algorithm should be clear and unambiguous. Each of its steps(or** phases), and their inputs/outputs should be clear and must lead to only one meaning.
- **Input - An algorithm should have 0 or more well-defined inputs.**
- **Output - An algorithm should have 1 or more well-defined outputs, and should** match the desired output.
- **Finiteness - Algorithms must terminate after a finite number of steps.**
- **Feasibility - Should be feasible with the available resources.**
- **Independent - An algorithm should have step-by-step directions , which should** be independent of any 92 2017 __programming code.___

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Advantages of Algorithm

- It is a step-by-step repetition of a solution to a given problem which is very easy to understand.
- Its easy to first develop an algorithm, then convert it into a flowchart &then into a computer program.
- Its is easy to debug as every step is got its own logical sequence.

Disadvantages of algorithm.

- Writing algorithm takes a long time.
- An Algorithm is not a computer program, it is rather a concept of how a program should be.

Example of Algorithm

- Let's try to learn algorithm-writing by using an example.
- **Problem- Design an algorithm to add two numbers**

- **Step 1 - START**
- **Step 2 - declare three integers a, b & c**
- **Step 3 - define values of a & b**
- **Step 4 - add values of a & b**
- **Step 5 - store output of step 4 to c**
- **Step 6 - print c**

Ston 7 - STOP

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- Algorithms tell the programmers how to code the program. Alternatively, the algorithm can be written as -

- **Step 1 - START ADD**
- **Step 2 - get values of a & b**
- **Step 3 - c <— a + b**
- **Step 4 - display c**
- **Step 5 - STOP**

Flowchart

- A flowchart is a picture (graphical representation) of the problem solving process.
- A flowchart gives a step-by-step procedure for solution of a problem.
- **Elements of a flowchart:**
- Various geometrical shaped boxes represent the steps of the solution.
- The boxes are connected by directional arrows to show the flow of the solution.
- **Uses of a flowchart:**
- To specify the method of solving a problem.
- To plan the sequence of a computer program.
- Communicate ideas, solutions.

r Advantages Of Using FLOWCHARTS

1 .Communication: Flowcharts are better way of communicating the logic of a system to all concerned or involved.
2. Effective analysis: With the help of flowchart, problem can be analyzed in more effective way therefore reducing cost and wastage of time.
3. Proper documentation: Program flowcharts serve as a good program documentation, which is needed for various purposes, making things more efficient.
4. Efficient Coding: The flowcharts act as a guide or blueprint during the systems analysis and program development pnase.
5. Proper Debugging: The flowchart helps in debugging process.'
6. Efficient Program Maintenance: The maintenance of operatinq proqram becomes easy with the help of
flowcharts, it help the programmer to put efforts more efficiently on that part _ ''' J

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Disadvantages Of Using FLOWCHARTS:

1. **Complex logic:** Sometimes, the program logic is quite complicated. In that case, flowchart becomes complex and clumsy. This will become a pain for the user, resulting in a waste of time and money trying to correct the problem.
2. **Alterations and Modifications:** If alterations are required the flowchart may require re-drawing completely. This will usually waste valuable time.
3. **Reproduction:** As the flowchart symbols cannot be typed, reproduction of flowchart becomes a problem.

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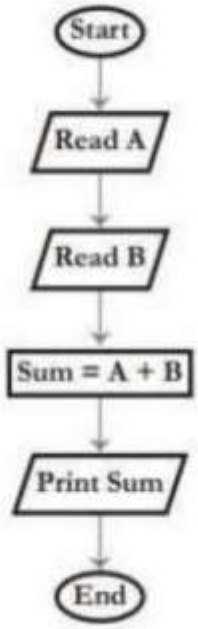
Flowchart symbols and their purpose

Flowchart symbols	Geometric shape	Purpose
Ellipse	o	Ellipse is used to indicate the start and end of a flowchart. Start wmten in the ellipse indicates die beginning of a flowchart, f ind or Stop or Exit written in the ellipse indicates the end of the flowchart.
Parallelogram	/ 7	A parallelogram is used to read ilata (input) or to pnnt data (output).
Rectangle		A rectangle is used to show the processing that takes place in the flowchart.
Diamond	No i	A diamond with two branches is used to show the decision making step in a flowchart. A question is specified m the diamond. The next step in the sequence is based on the answer to the question which is "Yes" or “No”.
Arrows	A > 4 H	Arrows are used to connect the steps in a flowchart, to show the flow or sequence of the problem solving process

Flowchart - How to find sum of two numbers



Finding sum of 845 and 247

Start

A= 845

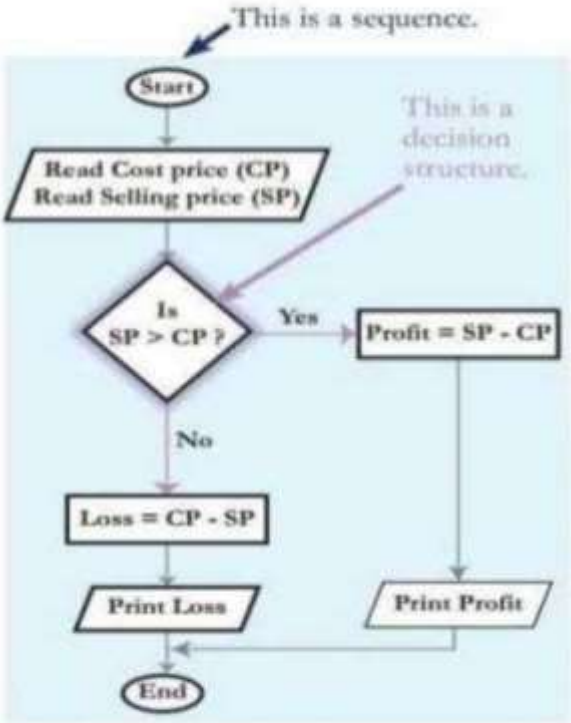
B= 247

Sum= 845+ 247

Sum= 1092

End

Flowchart - How to find profit or loss.



Finding profit or loss when CP = 325 and SP = 458

Start

Read CP=325
Read SP= 458

Condition:
Is 458 > 325? — Profit= 458-325

Profit= Rs. 133

End